Joe DiGiovanni

Data Scientist - AI Systems

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Summary

Data Scientist with an M.S. in Applied Data Science from UChicago. Focused on finding structure in ambiguous, high-variance problems and building systems to translate complex analytical outputs into practical insights. Experience spans causal inference research, LLM-powered product development, and end-to-end pipeline design.

Experience

Founder & Lead Data Scientist **Sep 2025 — Present**
FullCount AI *Remote*

- Built a public analytics platform querying over two-million rows of high dimensional Statcast data enabling non-technical users to create customized interactive visualizations.
- Diagnosed a core reliability failure in LLM-driven data querying of hallucinated SQL results and re-architected the system around a deterministic DuckDB routing layer, establishing correctness as a hard constraint rather than a best-effort outcome.
- Conducted structured discovery with alpha testers to identify trust and consistency failures in existing AI tools, replacing raw LLM output with modular UI components to make results verifiable and repeatable.

AI Pipeline Engineer **Mar 2025 — Aug 2025**
PowerKiosk, via UChicago Industry Capstone *Remote*

- Partnered with a digital energy marketplace to lead end-to-end system design for a 4-person team, delivering a solution that earned the highest recognition for impact and execution quality across the UChicago capstone cohort (awarded Best In Show).
- Transformed 1,000+ digital and scanned energy bills and contracts with no consistent schema into a fully queryable analytical layer by designing a multi-stage pipeline combining OCR, NLP information extraction, and vector embeddings.
- Replaced a manual document review process taking hours per query with a RAG-powered natural language interface returning results in seconds for energy market analysts.

Analytics Research Assistant **Sep 2023 — June 2024**
UCSB Baseball, University of California, Santa Barbara (NCAA D1) *Santa Barbara, CA*

- Built a predictive evaluation framework across 750+ players for a Division-1 program, isolating high-ROI recruiting targets from small-sample noise to inform coaching staff roster decisions.
- Synthesized video and radar tracking data into opponent scouting reports on high-leverage tendencies, becoming a repeatable input to game preparation.
- Modeled team bunting through Expected Value analysis to quantify run-scoring impact by game situation, turning coaching intuition into a data-backed framework that challenged prior in-game strategy.

Projects

Independent Quantitative Research on MLB Pitch Tunneling **Jun 2025 — Present**
The University of Chicago *Remote*

- Led an end-to-end independent research project with no established feature framework, building one from first principles and testing competing spatiotemporal feature specifications to achieve a 150%+ improvement in model R^2 for predicting swinging strike rate while preserving interpretability.
- Applied causal inference techniques to identify the primary mechanism driving observed outcomes, demonstrating a measurable increase (up to +23.4%) in target behavioral responses within strategically defined regions of the feature space
- Created interpretable visualizations translating complex statistical outputs from research experiments into clear, actionable insights for non-technical stakeholders while preparing findings for submission to a peer-reviewed journal.

Skills

AI & Machine Learning: LLMs (RAG, Agents), NLP, PyTorch, Scikit-learn, XGBoost, Computer Vision, Predictive Modeling
Statistics: Causal Inference, Hypothesis Testing, Regression, Experimental Design, Feature Engineering
Languages & Data: Python (pandas, NumPy), SQL, R, JavaScript, DuckDB, ETL Pipelines, REST APIs, Git
Visualization & Deployment: Tableau, Streamlit, Matplotlib, ggplot2, FastAPI, React, Docker, GitHub Actions, GCP, AWS

Education

Master of Science, Applied Data Science, *The University of Chicago* **Aug 2024 — Aug 2025**
Bachelor of Science, Applied Mathematics, *University of California, Santa Barbara* **Sep 2020 — June 2024**